

OSHIMA, Japan

WMO: 476750

Lat: 34.7494N

Lon: 139.3631E

Elev: 76

StdP: 100.42

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.3	2.5	-7.7	2.0	5.3	-5.9	2.3	5.6	14.2	8.1	13.3	8.4	4.0	80	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	30.5	26.2	29.7	25.8	28.9	25.4	27.0	29.1	26.5	28.6	26.0	28.0	5.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	22.0	28.4	25.8	21.3	28.0	25.3	20.6	27.5	85.5	29.2	83.2	28.6	80.9	28.2	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	11.1	9.7		-1.5	32.1	1.0	0.7	-2.2	32.5	-2.7	32.9	-3.3	33.3	-4.0	33.8
			DB	-3.2	27.7	1.1	0.7	-4.1	28.2	-4.7	28.6	-5.4	29.1	-6.2	29.6
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	7.3	7.6	10.3	14.4	18.3	21.2	25.0	26.4	23.6	19.2	14.4	9.9
	DBStd	6.93	2.23	2.62	2.92	2.78	2.04	1.96	2.16	1.75	2.50	2.43	2.62	2.65
	HDD10.0	233	88	75	32	3	0	0	0	0	0	0	1	34
	HDD18.3	1436	341	299	248	121	25	2	0	1	18	119	262	
	CDD10.0	2617	5	9	42	134	257	335	466	509	409	284	134	32
	CDD18.3	777	0	0	0	2	24	87	208	251	159	44	2	0
	CDH23.3	4846	0	0	0	0	18	178	1452	2208	900	89	0	0
	CDH26.7	942	0	0	0	0	0	8	271	522	138	3	0	0
Wind	WSAvg	4.9	5.4	5.2	5.5	5.2	4.7	4.3	4.5	4.3	4.6	5.0	5.1	5.5
Precipitation	PrecAvg	2560	124	147	213	223	237	293	223	195	301	293	198	114
	PrecMax	3275	306	272	419	403	475	522	567	510	499	843	535	217
	PrecMin	1710	9	27	75	94	91	132	6	4	90	57	59	9
	PrecStd	384	73	73	80	82	93	116	140	132	114	167	115	57
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	17.4	19.3	22.2	25.1	27.9	31.2	31.6	30.3	27.0	22.4	19.6
		MCWB	13.3	14.4	14.9	17.3	19.6	23.8	26.1	26.9	25.8	22.5	18.6	17.3
	2%	DB	14.3	15.4	17.8	20.9	23.7	26.1	30.1	30.7	29.2	25.5	21.1	17.3
		MCWB	11.0	12.7	14.3	17.2	19.2	22.5	25.9	26.4	25.3	21.8	17.5	14.2
	5%	DB	12.8	13.9	16.7	19.9	22.8	25.1	29.2	30.0	28.3	24.2	20.0	15.9
		MCWB	9.0	10.5	13.7	16.5	18.8	21.9	25.5	26.1	24.8	21.2	16.8	12.0
	10%	DB	11.4	12.5	15.7	18.9	22.0	24.3	28.2	29.3	27.3	23.1	18.9	14.4
		MCWB	7.6	8.9	12.5	15.5	18.4	21.5	25.1	25.8	24.4	20.0	15.6	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	15.6	16.8	20.1	22.2	24.8	27.3	27.9	26.7	24.1	20.2	18.2
		MCDB	15.4	16.6	17.6	20.6	23.0	26.1	29.6	30.2	28.8	25.5	21.3	18.9
	2%	WB	11.7	13.3	15.5	18.5	21.0	23.8	26.5	27.2	26.1	22.9	18.8	14.9
		MCDB	13.8	14.8	17.1	19.8	22.2	24.9	28.8	29.4	28.1	24.6	20.3	16.7
	5%	WB	9.6	11.2	14.3	17.4	20.1	23.1	26.0	26.8	25.5	21.7	17.4	12.7
		MCDB	12.1	13.6	16.3	19.1	21.7	24.3	28.2	28.8	27.4	23.8	19.3	15.0
	10%	WB	8.0	9.3	12.9	16.2	19.2	22.3	25.5	26.3	24.9	20.5	16.3	11.2
		MCDB	11.0	11.8	15.2	18.3	21.1	23.7	27.6	28.4	26.8	22.6	18.3	13.6
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.7	6.8	6.8	6.3	5.2	4.9	5.2	5.2	5.2	5.8	6.2
		MCDBR	7.7	7.9	7.7	7.5	7.5	6.3	5.9	5.9	6.3	6.4	7.0	7.5
		MCWBR	5.9	6.5	6.2	5.1	4.7	3.7	3.0	3.1	3.5	4.3	5.1	5.7
	5% WB	MCDBR	7.2	7.4	7.0	6.0	5.9	4.7	5.3	5.4	5.7	5.8	6.3	7.0
		MCWBR	6.4	7.1	6.7	5.2	4.7	3.2	2.9	3.1	3.5	4.6	5.5	6.2
Clear-Sky Solar Irradiance	taub	0.382	0.430	0.488	0.532	0.544	0.560	0.555	0.540	0.493	0.478	0.425	0.382	
	taud	2.255	2.119	1.976	1.912	1.947	1.988	2.048	2.082	2.183	2.131	2.202	2.263	
	Ebn at Noon	801	797	782	769	764	750	751	754	771	740	744	773	
	Edh at Noon	107	137	171	192	187	180	168	160	138	134	113	101	
All-Sky Solar Radiation	RadAvg	2.63	3.18	4.03	4.73	5.15	4.50	5.20	5.26	4.08	3.08	2.50	2.36	
	RadStd	0.20	0.34	0.27	0.51	0.55	0.45	0.81	0.43	0.35	0.32	0.22	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page